

Q.1 (15) $f_1(t) = U(t+1) - U(t-1)$, and $f_2(t) = U(t-1) - U(t-3)$.

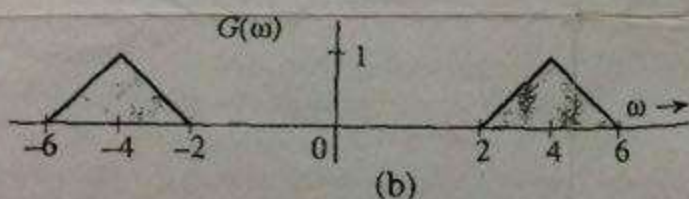
a) Compute $g(t) = f_1(t) * f_2(t)$ by applying time-shift property of convolution and using the fact that $U(t) * U(t) = tU(t)$.

b) Plot $g(t)$ for $-\infty < t < \infty$.

Q.2 (20): Compute the Fourier transform $X(\omega)$ of the following function;

$$x(t) = \begin{cases} \cos \frac{\pi}{2} t, & -1 < t \leq 1 \\ 0, & \text{otherwise.} \end{cases}$$

Q.3 (15): Find the inverse Fourier transform $g(t)$ of the following function;



(b)

Useful Tables

a) Fourier Transforms

$$\Delta\left(\frac{t}{\tau}\right) \Leftrightarrow \frac{\tau}{2} \text{sinc}^2\left(\frac{\omega\tau}{4}\right)$$

$$\text{rect}\left(\frac{t}{\tau}\right) \Leftrightarrow \tau \text{sinc}\left(\frac{\omega\tau}{2}\right)$$

b) Fourier Transform Properties: If $x(t) \Leftrightarrow X(\omega)$, then

$$X(t) \Leftrightarrow 2\pi x(-\omega)$$

$$x(at) \Leftrightarrow \frac{1}{|a|} X\left(\frac{\omega}{a}\right)$$

$$x(t-t_0) \Leftrightarrow e^{-j\omega t_0} X(\omega)$$

$$x(t)e^{j\omega_0 t} \Leftrightarrow X(\omega - \omega_0)$$

$$x_1(t) * x_2(t) \Leftrightarrow X_1(\omega) X_2(\omega)$$

$$x_1(t)x_2(t) \Leftrightarrow \frac{1}{2\pi} X_1(\omega) * X_2(\omega)$$

$$2t \int x(t) * x(t-\tau) d(\tau) \int U(t+1) U(t-1) dt$$

$$= t U(t-1)$$